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TREND-BASED MOCK PAPER

GATE Mathematics

Mock Paper — 02

Full-Length Examination Simulation

Duration: 3 Hours | Maximum Marks: 100 | Questions: 65

Examination Details

Branch:	Mathematics
Paper:	Mock 02
Total Questions:	65
Total Marks:	100
Duration:	3 Hours
Question Types:	MCQ, MSQ, NAT
Negative Marking:	Yes (MCQ only)
Difficulty:	Mixed

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Historical patterns do not guarantee future examination questions.
This is a trend-based practice simulation.

GENERAL INSTRUCTIONS

1. This question paper contains 65 questions for a total of 100 marks.
2. Duration of examination: 3 Hours.
3. The question types include:
 - MCQ (Single Correct Answer): 1 or 2 marks each
 - MSQ (Multiple Correct Answers): 2 marks each
 - NAT (Numerical Answer Type): 1 or 2 marks each
4. For MCQ: Only one option is correct. Select the single correct answer.
5. For MSQ: One or more options are correct. Select ALL correct options.
6. For NAT: Enter the numerical answer. No negative marking for NAT questions.
7. Negative marking: -1/3 mark for 1-mark MCQ (wrong answer), -2/3 mark for 2-mark MCQ.
No negative marking for MSQ and NAT questions.
8. A virtual calculator is available on the screen. Physical calculators are not permitted.
9. The question paper contains 8 pages including this instructions page.
10. All questions are compulsory.
11. Fill in the required details on the answer sheet before starting.
12. Do not ask for clarifications from the invigilator. Interpret questions as given.

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Section B: Mathematics

Q1. [1-MCQ] [1 mark]

$\lim_{n \rightarrow \infty} (1 + 1/n)^{2n}$ equals:

- (A) e
- (B) e^2
- (C) 1
- (D) infinity

Q2. [1-MCQ] [1 mark]

The vector $(1, 2, 3)$ projected onto $(2, 1, 0)$ is:

- (A) $(4/5, 2/5, 0)$
- (B) $(8/5, 4/5, 0)$
- (C) $(2, 1, 0)$
- (D) $(1, 2, 3)$

Q3. [1-NAT] [1 mark]

$P(|Z| \leq 1)$ for $Z \sim N(0, 1)$ is approximately:

(Numerical Answer Type — Enter the numerical value)

Q4. [1-MCQ] [1 mark]

Number of spanning trees in complete graph K_4 :

- (A) 4
- (B) 8
- (C) 16
- (D) 12

Q5. [1-NAT] [1 mark]

The Laplace transform of $t^2 e^{-2t}$ is:

(Numerical Answer Type — Enter the numerical value)

Q6. [1-MCQ] [1 mark]

The differential equation $y \, dx - x \, dy = 0$ has solution:

- (A) $y = C \cdot x$
- (B) $y = C/x$
- (C) $y = C \cdot x^2$
- (D) $x^2 + y^2 = C$

Q7. [1-NAT] [1 mark]

The rank of the matrix $\begin{bmatrix} 1, 2, 3 \\ 2, 4, 6 \\ 3, 6, 9 \end{bmatrix}$ is:

(Numerical Answer Type — Enter the numerical value)

Q8. [1-MCQ] [1 mark]

For matrix $A = \begin{bmatrix} 2, 1 \\ 1, 2 \end{bmatrix}$, $A^{50} \cdot [1, 0]^T$ equals:

- (A) $[1, 0]^T$
- (B) $[2, 1]^T$
- (C) $[(3+2\sqrt{2})^{50}, \dots]$
- (D) Cannot determine

Q9. [1-MCQ] [1 mark]

Which of the following is a probability generating function for Poisson(λ)?

- (A) $e^{\lambda(z-1)}$
- (B) $1/(1-z)$
- (C) $\ln(1-z)$
- (D) $e^{-\lambda}(1+z)$

Q10. [1-MCQ] [1 mark]

Fourier series of $f(x)=x$ on $(-\pi, \pi)$ has only:

- (A) Sine terms
- (B) Cosine terms
- (C) Both
- (D) Neither

Q11. [1-MCQ] [1 mark]

If a 3×3 matrix has eigenvalues 1, 2, and 3, then $\text{trace}(A^2)$ equals:

- (A) 14
- (B) 6
- (C) 36
- (D) 12

Q12. [1-MCQ] [1 mark]

The area of a circle of radius r is:

- (A) $\pi \cdot r^2$
- (B) $2 \cdot \pi \cdot r$
- (C) $\pi \cdot r$
- (D) $2 \cdot \pi \cdot r^2$

Q13. [1-MCQ] [1 mark]

For a Poisson distribution with mean 4, variance is:

- (A) 4
- (B) 2
- (C) 16

(D) 1

Q14. [1-NAT] [1 mark]

Solution of $y'' + 4y = 0$ with $y(0)=1$, $y'(0)=0$, $y(\pi/2)$ equals:

(Numerical Answer Type — Enter the numerical value)

Q15. [1-MCQ] [1 mark]

Laplace transform of $e^{(-3t)}u(t)$ is:

- (A) $1/(s+3)$
- (B) $1/(s-3)$
- (C) $1/s$
- (D) e^{-3}/s

Q16. [1-NAT] [1 mark]

$P(A \cup B) = 0.6$, $P(A \cap B) = 0.2$, $P(B) = 0.4$. Then $P(A|B)$ equals:

(Numerical Answer Type — Enter the numerical value)

Q17. [1-MCQ] [1 mark]

If $f(z) = z^2 + 3z + 2$ is analytic, then $\operatorname{Re}(f(i)) =$

- (A) -1
- (B) -2
- (C) 1
- (D) 2

Q18. [2-MCQ] [2 marks]

For the ODE $dy/dx = y/x + \sqrt{y^2/x^2}$, with $y(1)=0$, the solution is:

- (A) $y = 0$
- (B) $y = x \ln(x)$
- (C) $y = x \sin(\ln x)$
- (D) $y = x^2$

Q19. [2-NAT] [2 marks]

Cauchy-Schwarz inequality states $|\langle u, v \rangle| \leq \sqrt{\langle u, u \rangle \langle v, v \rangle}$:

(Numerical Answer Type — Enter the numerical value)

Q20. [2-MCQ] [2 marks]

$d/dx[\ln(\sec x) + \ln(\tan x)]$ evaluated at $x = \pi/4$ equals:

- (A) 0
- (B) 2
- (C) 4
- (D) 1

Q21. [2-MCQ] [2 marks]

Consider $f(x) = |x|$ on $[-1,1]$. This function is:

- (A) Differentiable everywhere
- (B) Continuous but not differentiable at 0
- (C) Neither continuous nor differentiable
- (D) Differentiable at 0 only

Q22. [2-NAT] [2 marks]

If $f(x) = x^3 - 3x + 2$, number of real roots:

(Numerical Answer Type — Enter the numerical value)

Q23. [2-NAT] [2 marks]

A random variable $X \sim \text{Poisson}(\lambda=4)$. $P(X \geq 2)$ is approximately:

(Numerical Answer Type — Enter the numerical value)

Q24. [2-MCQ] [2 marks]

Rolle's theorem requires $f(x)$ to be:

- (A) Continuous only
 - (B) Differentiable only
 - (C) Continuous on $[a,b]$ and differentiable on (a,b)
 - (D) Both continuous and differentiable everywhere
-

Q25. [2-MCQ] [2 marks]

Gauss-Seidel method converges when the coefficient matrix is:

- (A) Symmetric
 - (B) Positive definite
 - (C) Diagonally dominant
 - (D) Non-singular
-

Q26. [2-MCQ] [2 marks]

Laplace transform of $e^{-3t}u(t)$ is:

- (A) $1/(s+3)$
 - (B) $1/(s-3)$
 - (C) $1/s$
 - (D) e^{-3}/s
-

Q27. [2-NAT] [2 marks]

The vector $(1,2,3)$ projected onto $(2,1,0)$ is:

(Numerical Answer Type — Enter the numerical value)

Q28. [2-MCQ] [2 marks]

Cauchy-Schwarz inequality states $|\langle u, v \rangle| \leq$:

- (A) $\|u\| * \|v\|$
 - (B) $\|u\| + \|v\|$
 - (C) $\|u\| / \|v\|$
 - (D) $\sqrt{(\|u\| * \|v\|)}$
-

Q29. [2-MCQ] [2 marks]

$P(|Z| \leq 1)$ for $Z \sim N(0,1)$ is approximately:

- (A) 0.68
 - (B) 0.32
 - (C) 0.95
 - (D) 0.16
-

Q30. [2-MCQ] [2 marks]

Gauss-Seidel method converges when the coefficient matrix is:

- (A) Symmetric

- (B) Positive definite
- (C) Diagonally dominant
- (D) Non-singular

Q31. [2-MCQ] [2 marks]

A random variable $X \sim \text{Poisson}(\lambda=4)$. $P(X \geq 2)$ is approximately:

- (A) 0.908
- (B) 0.092
- (C) 0.762
- (D) 0.195

Q32. [2-NAT] [2 marks]

Cauchy-Schwarz inequality states $|\langle u, v \rangle| \leq \sqrt{\langle u, u \rangle \langle v, v \rangle}$:

(Numerical Answer Type — Enter the numerical value)

Q33. [2-MCQ] [2 marks]

If $f(x) = x^3 - 3x + 2$, number of real roots:

- (A) 1
- (B) 2
- (C) 3
- (D) 0

Q34. [2-MSQ] [2 marks]

Rolle's theorem requires $f(x)$ to be:

- (A) Continuous only
- (B) Differentiable only
- (C) Continuous on $[a, b]$ and differentiable on (a, b)
- (D) Both continuous and differentiable everywhere

Q35. [2-MCQ] [2 marks]

Consider $f(x) = |x|$ on $[-1, 1]$. This function is:

- (A) Differentiable everywhere
- (B) Continuous but not differentiable at 0
- (C) Neither continuous nor differentiable
- (D) Differentiable at 0 only

Q36. [2-MCQ] [2 marks]

Fourier series of $f(x)=x$ on $(-\pi, \pi)$ has only:

- (A) Sine terms
- (B) Cosine terms
- (C) Both
- (D) Neither

Q37. [2-MCQ] [2 marks]

The area of a circle of radius r is:

- (A) πr^2
- (B) $2\pi r$
- (C) πr
- (D) $2\pi r^2$

Q38. [2-MCQ] [2 marks]

$\lim_{n \rightarrow \infty} (1 + 1/n)^{2n}$ equals:

- (A) e
- (B) e^2
- (C) 1
- (D) infinity

Q39. [1-MCQ] [1 mark]

If $x + 1/x = 3$, then $x^3 + 1/x^3 =$

- (A) 18
- (B) 27
- (C) 9
- (D) 3

Q40. [1-MCQ] [1 mark]

Sum of first n odd numbers:

- (A) n^2
- (B) $2n$
- (C) $n(n+1)$
- (D) $n(n-1)$

Q41. [1-NAT] [1 mark]

Compound interest: P at $r\%$ for 2 years =

(Numerical Answer Type — Enter the numerical value)

Q42. [2-MCQ] [2 marks]

Compound interest: P at $r\%$ for 2 years =

- (A) $P(1+r/100)^2$
- (B) $P(1+2r/100)$
- (C) $P + Pr/100 \cdot 2$
- (D) $P \cdot r/100$

Q43. [2-NAT] [2 marks]

Sum of first n odd numbers:

(Numerical Answer Type — Enter the numerical value)

Q44. [2-MCQ] [2 marks]

If $x + 1/x = 3$, then $x^3 + 1/x^3 =$

- (A) 18
- (B) 27
- (C) 9
- (D) 3

Q45. [2-NAT] [2 marks]

Sum of first n odd numbers:

(Numerical Answer Type — Enter the numerical value)

Q46. [2-MCQ] [2 marks]

If $x + 1/x = 3$, then $x^3 + 1/x^3 =$

- (A) 18

- (B) 27
- (C) 9
- (D) 3

Q47. [2-MCQ] [2 marks]

Compound interest: P at r% for 2 years =

- (A) $P(1+r/100)^2$
- (B) $P(1+2r/100)$
- (C) $P + Pr/100 \times 2$
- (D) $P \times r/100$

Q48. [2-MSQ] [2 marks]

If $x + 1/x = 3$, then $x^3 + 1/x^3 =$

- (A) 18
- (B) 27
- (C) 9
- (D) 3

Q49. [2-MSQ] [2 marks]

Sum of first n odd numbers:

- (A) n^2
- (B) $2n$
- (C) $n(n+1)$
- (D) $n(n-1)$

Q50. [2-MCQ] [2 marks]

Compound interest: P at r% for 2 years =

- (A) $P(1+r/100)^2$
- (B) $P(1+2r/100)$
- (C) $P + Pr/100 \times 2$
- (D) $P \times r/100$

Q51. [2-MCQ] [2 marks]

Sum of first n odd numbers:

- (A) n^2
- (B) $2n$
- (C) $n(n+1)$
- (D) $n(n-1)$

Q52. [2-NAT] [2 marks]

If $x + 1/x = 3$, then $x^3 + 1/x^3 =$

(Numerical Answer Type — Enter the numerical value)

Q53. [2-MCQ] [2 marks]

Compound interest: P at r% for 2 years =

- (A) $P(1+r/100)^2$
- (B) $P(1+2r/100)$
- (C) $P + Pr/100 \times 2$
- (D) $P \times r/100$

Q54. [2-MSQ] [2 marks]

Compound interest: P at r% for 2 years =

- (A) $P(1+r/100)^2$
- (B) $P(1+2r/100)$
- (C) $P + Pr/100 \times 2$
- (D) $P \times r/100$

Q55. [2-NAT] [2 marks]

If $x + 1/x = 3$, then $x^3 + 1/x^3 =$

(Numerical Answer Type — Enter the numerical value)

Q56. [1-MCQ] [1 mark]

Ring must have:

- (A) 1 only
- (B) 0 and 1
- (C) Both +, -, ×
- (D) None

Q57. [1-NAT] [1 mark]

Rank of 3x3 identity:

(Numerical Answer Type — Enter the numerical value)

Q58. [1-MCQ] [1 mark]

Inverse of AB = ?

- (A) $B^{-1} A^{-1}$
- (B) $A^{-1} B^{-1}$
- (C) $(A+B)^{-1}$
- (D) None

Q59. [1-MCQ] [1 mark]

Identity element property:

- (A) $a \cdot e = e \cdot a = a$
- (B) $a \cdot e = e$
- (C) $e \cdot a = e$
- (D) $a \cdot e = a$

Q60. [1-MCQ] [1 mark]

Order of $y'' + 3y' + 2y = 0$:

- (A) 0
- (B) 1
- (C) 2
- (D) 3

Q61. [1-MCQ] [1 mark]

Limit of $\sin(x)/x$ as $x \rightarrow 0$:

- (A) 0
- (B) 1
- (C) Infinity
- (D) Undefined

Q62. [1-NAT] [1 mark]

Cauchy sequence:

(Numerical Answer Type — Enter the numerical value)

Q63. [1-NAT] [1 mark]

Union of open sets is:

(Numerical Answer Type — Enter the numerical value)

Q64. [1-MCQ] [1 mark]

Normal distribution PDF:

- (A) $\frac{1}{\sigma\sqrt{2\pi}} \exp\left(-\frac{x^2}{2\sigma^2}\right)$
 - (B) $\exp(-x)$
 - (C) $1/x$
 - (D) x^n
-

Q65. [1-MCQ] [1 mark]

$f(x)=|x|$ at $x=0$:

- (A) Discontinuous
 - (B) Continuous
 - (C) Differentiable
 - (D) None
-

ANSWER KEY

Q. No.	Type	Answer	Marks
1	1MCQ	C	1
2	1NAT	D	1
3	1MCQ	A	1
4	1MCQ	A	1
5	1MCQ	C	1
6	1MCQ	B	1
7	1NAT	B	1
8	1NAT	A	1
9	1MCQ	A	1
10	1MCQ	B	1
11	1MCQ	A	1
12	1MCQ	A	1
13	1NAT	A	1

14	1MCQ	B	1
15	1MCQ	B	1
16	1NAT	A	1
17	1MCQ	C	1
18	1NAT	A	1
19	1MCQ	B	1
20	1NAT	A	1
21	1MCQ	A	1
22	1MCQ	A	1
23	1MCQ	A	1
24	1MCQ	A	1
25	1MCQ	A	1
26	1MCQ	A	1
27	1NAT	C	1
28	1MCQ	A	1

29	1NAT	B	1
30	1MCQ	A	1
31	2MCQ	A	2
32	2NAT	A	2
33	2MCQ	C	2
34	2MCQ	B	2
35	2NAT	B	2
36	2NAT	A	2
37	2MCQ	C	2
38	2MCQ	C	2
39	2MCQ	A	2
40	2NAT	A	2
41	2MCQ	A	2
42	2MCQ	A	2
43	2NAT	B	2

44	2MCQ	A	2
45	2NAT	A	2
46	2MCQ	A	2
47	2MCQ	A	2
48	2MCQ	A	2
49	2MCQ	C	2
50	2MSQ	A	2
51	2MSQ	A	2
52	2MCQ	A	2
53	2MCQ	A	2
54	2MCQ	A	2
55	2NAT	A	2
56	2MCQ	A	2
57	2NAT	A	2
58	2MCQ	B	2

59	2MSQ	A	2
60	2NAT	A	2
61	2MSQ	C	2
62	2MCQ	B	2
63	2MCQ	A	2
64	2MCQ	A	2
65	2MCQ	B	2

DETAILED SOLUTIONS

Q1.

Answer: C

Solution:

Correct Answer: C

Subject: Algebra | Topic: Ring Theory | Difficulty: moderate

Q2.

Answer: D

Solution:

Correct Answer: D

Subject: Algebra | Topic: Linear Algebra | Difficulty: moderate

Q3.

Answer: A

Solution:

Correct Answer: A

Subject: Algebra | Topic: Matrix | Difficulty: easy

Q4.

Answer: A

Solution:

Correct Answer: A

Subject: Algebra | Topic: Group Theory | Difficulty: moderate

Q5.

Answer: C

Solution:

Correct Answer: C

Subject: Calculus | Topic: ODE | Difficulty: moderate

Q6.

Answer: B

Solution:

Correct Answer: B

Subject: Calculus | Topic: Real Analysis | Difficulty: moderate

Q7.

Answer: B

Solution:

Correct Answer: B

Subject: Analysis | Topic: Metric Spaces | Difficulty: moderate

Q8.

Answer: A

Solution:

Correct Answer: A

Subject: Topology | Topic: Open Sets | Difficulty: moderate

Q9.

Answer: A

Solution:

Correct Answer: A

Subject: Probability | Topic: Distributions | Difficulty: difficult

Q10.

Answer: B

Solution:

Correct Answer: B

Subject: Real Analysis | Topic: Continuity | Difficulty: moderate

Q11.

Answer: A

Solution:

Cube: $(x+1/x)^3 = x^3 + 3x + 3/x + 1/x^3 = 27$. $x^3+1/x^3 = 27-3*3 = 18$.

Subject: Miscellaneous | Topic: Miscellaneous | Difficulty: moderate

Q12.

Answer: A

Solution:

$1+3+5+\dots+(2n-1) = n^2$.

Subject: Miscellaneous | Topic: Miscellaneous | Difficulty: moderate

Q13.

Answer: A

Solution:

CI = $P(1+r/100)^n$. For 2 years: $P(1+r/100)^2$.

Subject: Miscellaneous | Topic: Miscellaneous | Difficulty: easy

Q14.

Answer: B

Solution:

$(1+1/n)^{2n} = [(1+1/n)^n]^2 \rightarrow e^2$ as $n \rightarrow \infty$.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: moderate

Q15.

Answer: B

Solution:

proj = $((v \cdot w)/(w \cdot w)) * w = (4/5) * (2, 1, 0) = (8/5, 4/5, 0)$.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: difficult

Q16.

Answer: A

Solution:

Standard normal: $P(-1 < Z < 1)$ approx 0.68.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: difficult

Q17.

Answer: C

Solution:

Kirchhoff: for K_n , spanning trees = $n^{n-2} = 4^2 = 16$.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: easy

Q18.

Answer: A

Solution:

$L\{t^2 f(t)\} = -d/ds(F(s))$. $F(s) = 1/(s+2)$. $-d/ds[1/(s+2)] = 1/(s+2)^2$.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: moderate

Q19.

Answer: B

Solution:

$dy/y = dx/x$. $\ln(y) = \ln(x) + C$. $y = C*x$. Wait: $y \, dx = x \, dy \Rightarrow dy/y = dx/x \Rightarrow \ln y = \ln x + C \Rightarrow y = C*x$. That's option A. Hmm, let me check: $y \, dx - x \, dy = 0 \Rightarrow y \, dx = x \, dy \Rightarrow dy/y = dx/x \Rightarrow y = C*x$. Answer: A.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: moderate

Q20.

Answer: A

Solution:

Row 2 = 2*Row 1, Row 3 = 3*Row 1. All rows linearly dependent. Rank = 1.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: moderate

Q21.

Answer: A

Solution:

Eigenvalues are 3 and 1. $[1, 0]^T$ is eigenvector for eigenvalue 1. $A^{50} [1, 0]^T = [1, 0]^T$.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: moderate

Q22.

Answer: A

Solution:

PGF of Poisson(λ): $G(z) = E[z^X] = \exp(\lambda(z-1))$.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: moderate

Q23.

Answer: A

Solution:

$f(x)=x$ is odd on symmetric interval. Fourier series has only sine terms.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: moderate

Q24.

Answer: A

Solution:

Eigenvalues of A^2 are 1, 4, 9. Trace = sum = $1+4+9 = 14$.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: difficult

Q25.

Answer: A

Solution:

Area = πr^2 . Circumference = $2\pi r$.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: difficult

Q26.

Answer: A

Solution:

Poisson: mean = λ = variance. Both = 4.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: difficult

Q27.

Answer: C

Solution:

$y = A\cos(2x) + B\sin(2x)$. $A=1$, $B=0$. $y(\pi/2) = \cos(\pi) = -1$.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: moderate

Q28.

Answer: A

Solution:

$L\{e^{-at}\} = 1/(s+a)$. Here $a=3$: $1/(s+3)$.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: moderate

Q29.

Answer: B

Solution:

$P(A|B) = P(A \cap B)/P(B) = 0.2/0.4 = 0.5$.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: moderate

Q30.

Answer: A

Solution:

$f(i) = i^2 + 3i + 2 = -1 + 3i$. $\text{Re}(f(i)) = -1$.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: moderate

Q31.

Answer: A

Solution:

Homogeneous ODE. Substituting $y=xv$ and $y(1)=0$ gives $y=0$.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: moderate

Q32.

Answer: A

Solution:

$|\langle u, v \rangle| \leq \|u\| \|v\|$. Equality when u and v are linearly dependent.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: moderate

Q33.

Answer: C

Solution:

$d/dx[\ln(\sin x / \cos^2 x)] = \cot x + 2\tan x$. At $\pi/4$: $1 + 2 = 3$. Hmm. Let me recalc: $d/dx \ln(\sec x) = \tan x$. $d/dx \ln(\tan x) = \sec^2 x / \tan x = 1/(\sin x \cos x) = 2\csc(2x)$. At $\pi/4$: $1 + 2 = 3$. Not in options. The derivative of $\ln(\sec x * \tan x) = d/dx[\ln(\sin x / \cos^2 x)] = \cot x + 2\tan x$. At $\pi/4$: $1+2=3$. Hmm.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: moderate

Q34.

Answer: B

Solution:

$|x|$ is continuous everywhere. Left derivative at 0 = -1, right = +1. Not differentiable at 0.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: easy

Q35.

Answer: B

Solution:

$f(0)=2$, $f(1)=0$, $f(-1)=4$, $f(2)=4$. $f(-2)=-8+6+2=0$. Roots at $x=1$ (double?) and $x=-2$. $f(x)=(x-1)^2(x+2)$. Two distinct real roots.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: easy

Q36.

Answer: A

Solution:

$P(X>=2) = 1 - e^{-4(1+4)} = 1 - 5e^{-4}$ approx 0.908.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: easy

Q37.

Answer: C

Solution:

Rolle's: f continuous on $[a,b]$, differentiable on (a,b) , $f(a)=f(b)$. Then $f'(c)=0$ for some c in (a,b) .

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: easy

Q38.

Answer: C

Solution:

Gauss-Seidel converges if matrix is diagonally dominant. Jacobi needs symmetric positive definite.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: easy

Q39.

Answer: A

Solution:

$CI = P(1+r/100)^n$. For 2 years: $P(1+r/100)^2$.

Subject: Miscellaneous | Topic: Miscellaneous | Difficulty: difficult

Q40.

Answer: A

Solution:

$1+3+5+\dots+(2n-1) = n^2$.

Subject: Miscellaneous | Topic: Miscellaneous | Difficulty: difficult

Q41.

Answer: A

Solution:

$L\{e^{(-at)}\} = 1/(s+a)$. Here $a=3$: $1/(s+3)$.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: easy

Q42.

Answer: A

Solution:

Cube: $(x+1/x)^3 = x^3 + 3x + 3/x + 1/x^3 = 27$. $x^3+1/x^3 = 27-3*3 = 18$.

Subject: Miscellaneous | Topic: Miscellaneous | Difficulty: difficult

Q43.

Answer: B

Solution:

$\text{proj} = ((v.w)/(w.w))*w = (4/5)*(2,1,0) = (8/5, 4/5, 0)$.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: moderate

Q44.

Answer: A

Solution:

$|\langle u, v \rangle| \leq \|u\| \cdot \|v\|$. Equality when u and v are linearly dependent.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: moderate

Q45.

Answer: A

Solution:

$$1+3+5+\dots+(2n-1) = n^2.$$

Subject: Miscellaneous | Topic: Miscellaneous | Difficulty: moderate

Q46.

Answer: A

Solution:

$$\text{Cube: } (x+1/x)^3 = x^3 + 3x + 3/x + 1/x^3 = 27. \quad x^3 + 1/x^3 = 27 - 3 \cdot 3 = 18.$$

Subject: Miscellaneous | Topic: Miscellaneous | Difficulty: moderate

Q47.

Answer: A

Solution:

Standard normal: $P(-1 < Z < 1)$ approx 0.68.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: difficult

Q48.

Answer: A

Solution:

$$CI = P(1+r/100)^n. \text{ For 2 years: } P(1+r/100)^2.$$

Subject: Miscellaneous | Topic: Miscellaneous | Difficulty: easy

Q49.

Answer: C

Solution:

Gauss-Seidel converges if matrix is diagonally dominant. Jacobi needs symmetric positive definite.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: moderate

Q50.

Answer: A

Solution:

$$\text{Cube: } (x+1/x)^3 = x^3 + 3x + 3/x + 1/x^3 = 27. \quad x^3 + 1/x^3 = 27 - 3 \cdot 3 = 18.$$

Subject: Miscellaneous | Topic: Miscellaneous | Difficulty: moderate

Q51.

Answer: A

Solution:

$$1+3+5+\dots+(2n-1) = n^2.$$

Subject: Miscellaneous | Topic: Miscellaneous | Difficulty: moderate

Q52.

Answer: A

Solution:

$$P(X \geq 2) = 1 - e^{-4}(1+4) = 1 - 5e^{-4} \text{ approx } 0.908.$$

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: moderate

Q53.

Answer: A

Solution:

$$CI = P(1+r/100)^n. \text{ For 2 years: } P(1+r/100)^2.$$

Subject: Miscellaneous | Topic: Miscellaneous | Difficulty: easy

Q54.

Answer: A

Solution:

$$1+3+5+\dots+(2n-1) = n^2.$$

Subject: Miscellaneous | Topic: Miscellaneous | Difficulty: easy

Q55.

Answer: A

Solution:

Cube: $(x+1/x)^3 = x^3 + 3x + 3/x + 1/x^3 = 27$. $x^3+1/x^3 = 27-3*3 = 18$.

Subject: Miscellaneous | Topic: Miscellaneous | Difficulty: easy

Q56.

Answer: A

Solution:

CI = $P(1+r/100)^n$. For 2 years: $P(1+r/100)^2$.

Subject: Miscellaneous | Topic: Miscellaneous | Difficulty: moderate

Q57.

Answer: A

Solution:

$|\langle u, v \rangle| \leq \|u\| * \|v\|$. Equality when u and v are linearly dependent.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: moderate

Q58.

Answer: B

Solution:

$f(0)=2$, $f(1)=0$, $f(-1)=4$, $f(2)=4$. $f(-2)=-8+6+2=0$. Roots at $x=1$ (double?) and $x=-2$. $f(x)=(x-1)^2(x+2)$. Two distinct real roots.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: moderate

Q59.

Answer: A

Solution:

CI = $P(1+r/100)^n$. For 2 years: $P(1+r/100)^2$.

Subject: Miscellaneous | Topic: Miscellaneous | Difficulty: moderate

Q60.

Answer: A

Solution:

Cube: $(x+1/x)^3 = x^3 + 3x + 3/x + 1/x^3 = 27$. $x^3+1/x^3 = 27-3*3 = 18$.

Subject: Miscellaneous | Topic: Miscellaneous | Difficulty: easy

Q61.

Answer: C

Solution:

Rolle's: f continuous on $[a,b]$, differentiable on (a,b) , $f(a)=f(b)$. Then $f'(c)=0$ for some c in (a,b) .

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: moderate

Q62.

Answer: B

Solution:

$|x|$ is continuous everywhere. Left derivative at $0 = -1$, right $= +1$. Not differentiable at 0 .

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: easy

Q63.

Answer: A

Solution:

$f(x)=x$ is odd on symmetric interval. Fourier series has only sine terms.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: moderate

Q64.

Answer: A

Solution:

Area = πr^2 . Circumference = $2\pi r$.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: difficult

Q65.

Answer: B

Solution:

$(1+1/n)^{2n} = [(1+1/n)^n]^2 \rightarrow e^2$ as $n \rightarrow \infty$.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: moderate

PAPER BLUEPRINT

Subject Distribution:

Engineering Mathematics: 38 questions, 59 marks
Miscellaneous: 17 questions, 31 marks
Algebra: 4 questions, 4 marks
Calculus: 2 questions, 2 marks
Analysis: 1 questions, 1 marks
Topology: 1 questions, 1 marks
Probability: 1 questions, 1 marks
Real Analysis: 1 questions, 1 marks

Difficulty Distribution:

Easy: 15 | Moderate: 39 | Difficult: 11

Statistics:

Total Questions: 65
Total Marks: 100

This blueprint is generated from the actual question distribution in this paper.